

## Experiment No. 6: Implement Page replacement algorithms using C

- a) Least Recently Used (LRU)
- b) Optimal

### a) Least Recently Used (LRU)

#### C PROGRAM

```
#include <stdio.h>
int main()
{
    int pages[50], frames[10], time[10];
    int n, f, i, j, k, flag, faults = 0, pos, min, counter = 0;

    printf("Enter number of pages: ");
    scanf("%d",&n);

    printf("Enter page reference string:\n");
    for(i=0;i<n;i++)
        scanf("%d",&pages[i]);

    printf("Enter number of frames: ");
    scanf("%d",&f);

    for(i=0;i<f;i++)
        frames[i] = -1;

    for(i=0;i<n;i++)
    {
        flag = 0; // ☒ fixed

        for(j=0;j<f;j++)
        {
            if(frames[j] == pages[i])
            {
                counter++;
                time[j] = counter;
                flag = 1; // ☒ fixed
                break;
            }
        }
    }
}
```

```

if(flag == 0)
{
    if(i < f)
    {
        frames[i] = pages[i];
        counter++;
        time[i] = counter;
        faults++;
    }
    else
    {
        min = time[0];
        pos = 0;

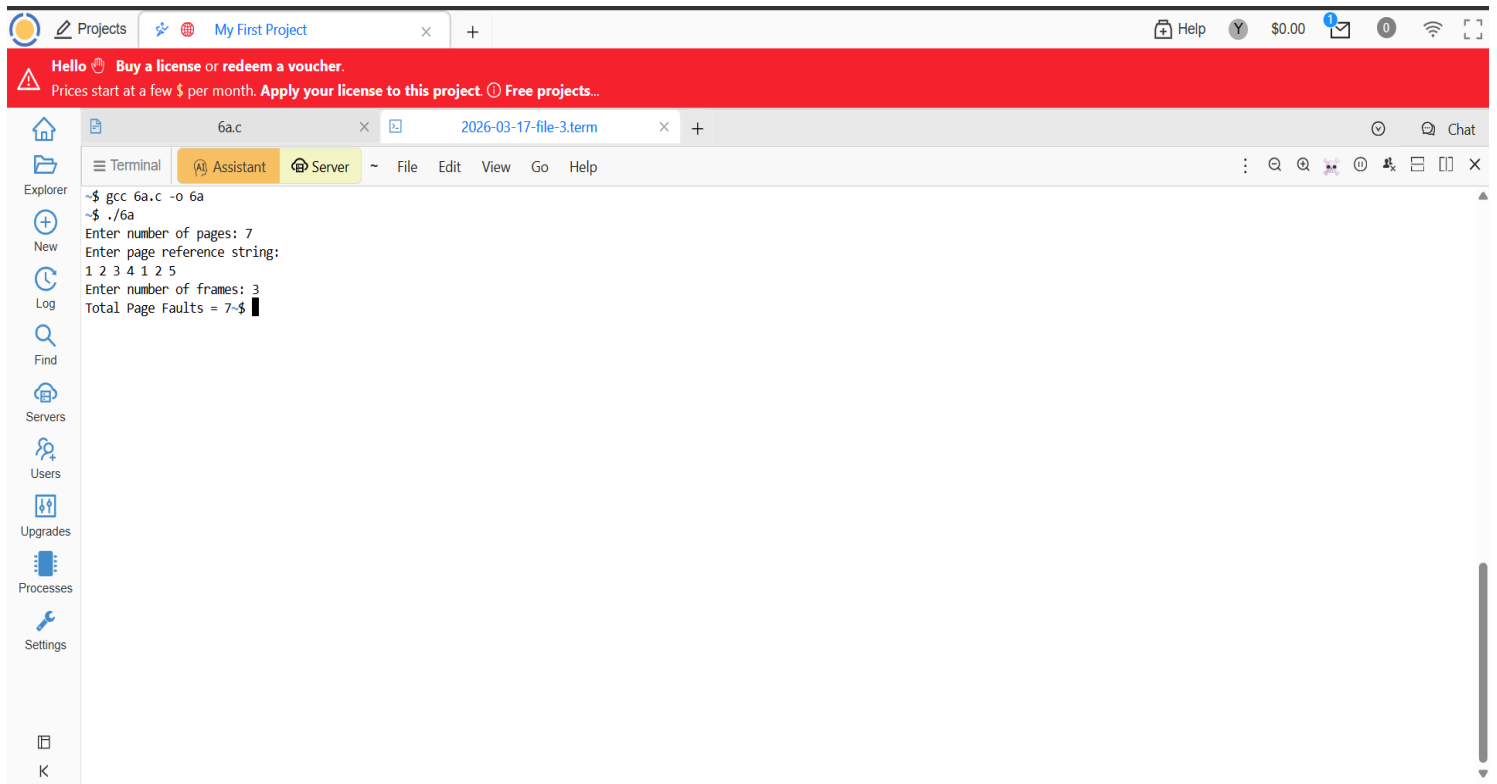
        for(k=1;k<f;k++)
        {
            if(time[k] < min)
            {
                min = time[k];
                pos = k;
            }
        }

        frames[pos] = pages[i];
        counter++;
        time[pos] = counter;
        faults++;
    }
}

printf("Total Page Faults = %d", faults);
return 0;
}

```

## OUTPUT :-



The screenshot shows a web-based IDE with a top navigation bar containing 'Projects' and 'My First Project'. A red banner at the top displays a message: 'Hello Buy a license or redeem a voucher. Prices start at a few \$ per month. Apply your license to this project. Free projects...'. The main workspace has a sidebar on the left with icons for Explorer, New, Log, Find, Servers, Users, Upgrades, Processes, and Settings. The central area contains a terminal window with the following output:

```
~$ gcc 6a.c -o 6a
~$ ./6a
Enter number of pages: 7
Enter page reference string:
1 2 3 4 1 2 5
Enter number of frames: 3
Total Page Faults = 7~$
```

## b) Optimal Page Replacement

### C PROGRAM

```
#include <stdio.h>
int main()
{
    int pages[50], frames[10];
    int n, f, i, j, k, flag, faults=0;
    int pos, farthest, index;
    printf("Enter number of pages: ");
    scanf("%d",&n);
    printf("Enter page reference string:\n");
    for(i=0;i<n;i++)
        scanf("%d",&pages[i]);

    printf("Enter number of frames: ");
    scanf("%d",&f);

    for(i=0;i<f;i++)
        frames[i] = -1;

    for(i=0;i<n;i++)
    {
        flag=0;

        for(j=0;j<f;j++)
        {
            if(frames[j]==pages[i])
            {
                flag=1;
                break;
            }
        }

        if(flag==0)
        {
            if(i<f)
            {
                frames[i]=pages[i];
                faults++;
            }
            else
            {
                farthest=-1;
            }
        }
    }
}
```

```

    for(j=0;j<f;j++)
    {
        index=-1;

        for(k=i+1;k<n;k++)
        {
            if(frames[j]==pages[k])
            {
                index=k;
                break;
            }
        }

        if(index== -1)
        {
            pos=j;
            break;
        }

        if(index>farthest)
        {
            farthest=index;
            pos=j;
        }
    }

    frames[pos]=pages[i];
    faults++;
}
}

printf("Total Page Faults = %d",faults);

return 0;
}

```

**OUTPUT :-**

Projects

My First Project

x

+

Help

Y

\$0.00

0

Hello Buy a license or redeem a voucher.

Prices start at a few \$ per month. Apply your license to this project. Free projects..

6a.c

x

6b.c

x

2026-03-17-file-4.term

x

+

Chat

Terminal

Assistant

Server

~

File

Edit

View

Go

Help

:

Q

Q

U

X

Explorer

New

Log

Find

Servers

Users

Upgrades

Processes

Settings

K

~\$ gcc 6b.c -o 6b

~\$ ./6b

Enter number of pages: 7

Enter page reference string:

7 0 1 2 0 3 0

Enter number of frames: 3

Total Page Faults = 5-\$